Data Warehousing for Partners: Data Warehouse Migration with BigQuery

Welcome to the 4-day Data Warehousing for Partners: Data Warehouse Migration with BigQuery course.

If you're wondering how to help your customers migrate Data Warehouse databases to BigQuery and move ETL pipelines to Google Cloud, then this deep dive technical course is for you. In this class, you will receive technical training for Enterprise Data Warehouses solutions using BigQuery based on the best practices developed internally by Google's technical sales and services organizations.

The course will also provide guidance and training on key technical challenges that can arise when migrating existing Enterprise Data Warehouses and ETL pipelines to Google Cloud. You will get hands-on experience with real migration tasks, such as data migration, schema optimization, and SQL Query conversion and optimization. The course will also cover key aspects of ETL pipeline migration to Dataproc as well as using Pub/Sub, Dataflow, and Cloud Data Fusion, giving you hands-on experience using all of these tools for Data Warehouse ETL pipelines.

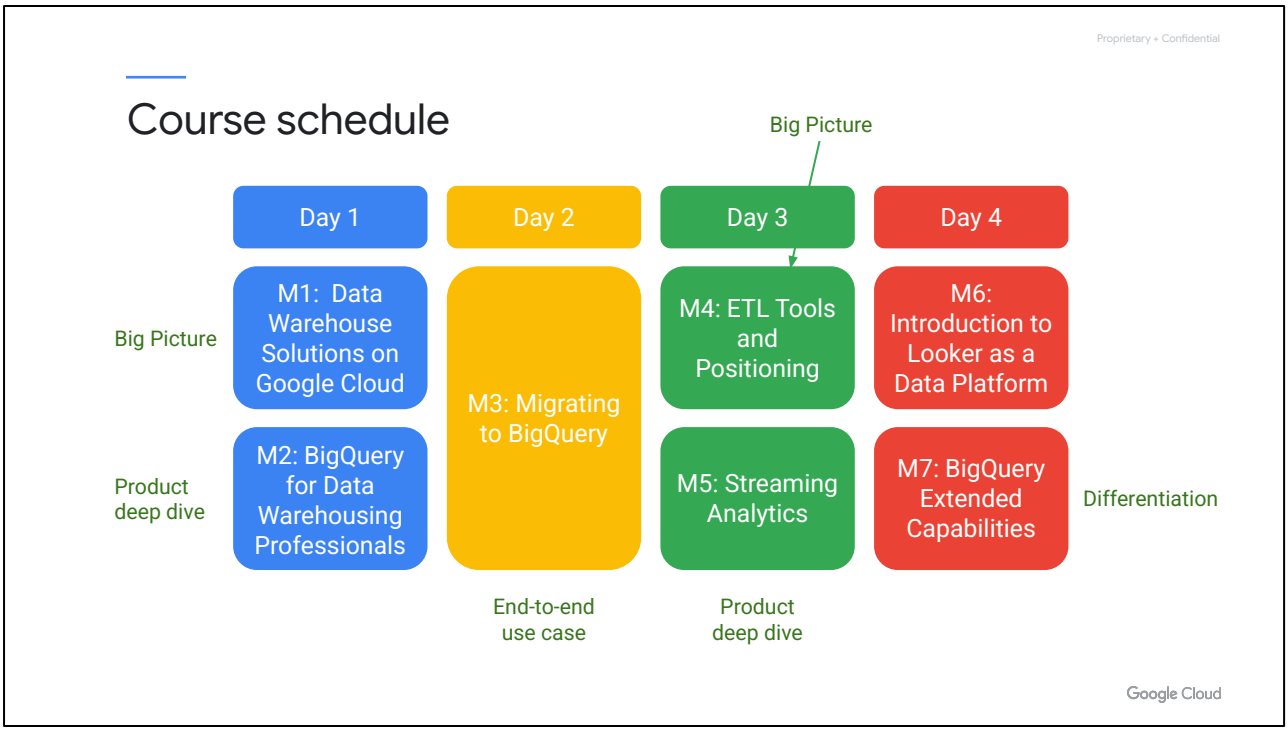
We are here in order to try to give you context and guidance on how to implement data warehousing on Google Cloud.

The training class is mainly focused around BigQuery – but there may be others solutions, including 3rd parties such as Snowflake who provide data warehousing solutions using Google Cloud. But our focus is BigQuery for now, in future we may expand.

Our objective is to introduce you to how to get started. Given that you will be working with customers to implement their data warehouses on Google Cloud, and the fact that they almost certainly already have an in-house on-premises Data Warehouse solution, your job will typically be focused on helping them migrate that to Google Cloud. Such migrations are complex so we want to show you what these migrations look like, give you insight into the process and access to tools that will help you successfully migrate and implement data warehouses to Google Cloud.

And remember that Data Warehousing isn't just about the data – it's also about upstream and downstream data pipelines and platforms. The ETL, or ELT, upstream data pipeline as well as the analytics/dashboards downstream.

So we will show you what you should be looking at on Google Cloud to fulfil these use cases.



The course will last four days.
There are seven modules:

- **Module 1:** Data Warehouse Solutions on Google Cloud
- **Module 2:** BigQuery for Data Warehousing Professionals
- **Module 3:** Migrating to BigQuery
- **Module 4:** ETL Tools and Positioning
- **Module 5:** Streaming Analytics
- **Module 6:** Introduction to Looker as a Data Platform
- **Module 7:** BigQuery Extended Capabilities

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We assume that you're familiar with data engineering on Google Cloud

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Course Agenda

Data Warehouse Solutions on Google Cloud

BigQuery for Data Warehousing Professionals

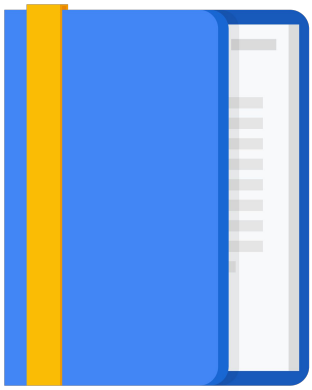
Migrating to BigQuery

ETL Tools and Positioning

Streaming Analytics

Introduction to Looker as a Data Platform

BigQuery Extended Capabilities



Google Cloud

The course is divided into 7 modules.

The first module discusses the key elements of Google's Data Warehouse solution portfolio and strategy.

The second module maps Enterprise Data Warehouse concepts and components to BigQuery and Google data services.

You'll then identify best practices for migrating data warehouses to BigQuery and demonstrate key skills required to perform successful migrations.

In the ETL Tools and Positioning module you will Implement data load and transformation pipelines for a BigQuery Data Warehouse using Google data processing and integration services.

In the Streaming Analytics module, you'll be able to describe how to implement a streaming analytics solution using Pub/Sub, Dataflow, and BigQuery.

In module 6, you'll be introduced to Looker and LookML to generate reports and gain insights.

And in the final module you'll explore the GIS, GIS Visualization, and Machine Learning enhancements to BigQuery.

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Facilities

Parking

Facilities

Food

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Course etiquette

Please silence your phone and take calls outside.

Recording this class is prohibited.

Ask questions interactively or via chat (online).

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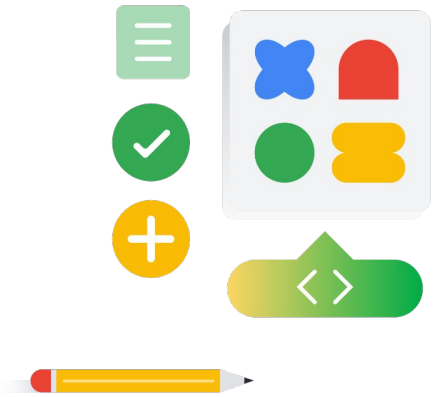
Lab environment

For each lab, Google Skills offers:

- A free set of resources for a fixed amount of time
- A clean environment with permissions



Google Cloud

Google Skills provisions you with Google account credentials, so you can access the Google Cloud Console for each lab at no cost.

Specifically, for each lab, Google Skills offers:

- A free set of resources for a fixed amount of time
- A clean environment with permissions

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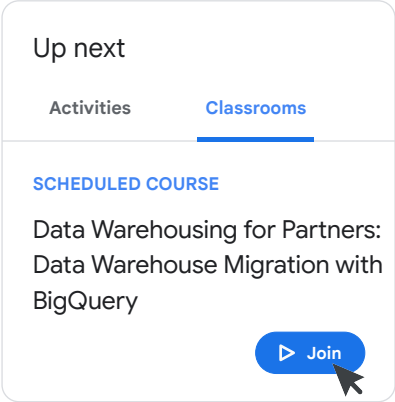
Open Google Skills

1. **Open an incognito window** (or private and/or anonymous window).
2. **Go** to the URL your instructor provides.
3. **Sign In** with existing account or **Join** with new account (with email you used to register for the course).
4. Launch the course from the **Dashboard** page.



Access issues

The process to open Google Skills can differ based on credentials used. Please reach out to your instructor if you have any access issues.

Google Cloud

Let's go ahead and open Google Skills.

1. **Open an incognito window** (or private/anonymous window).
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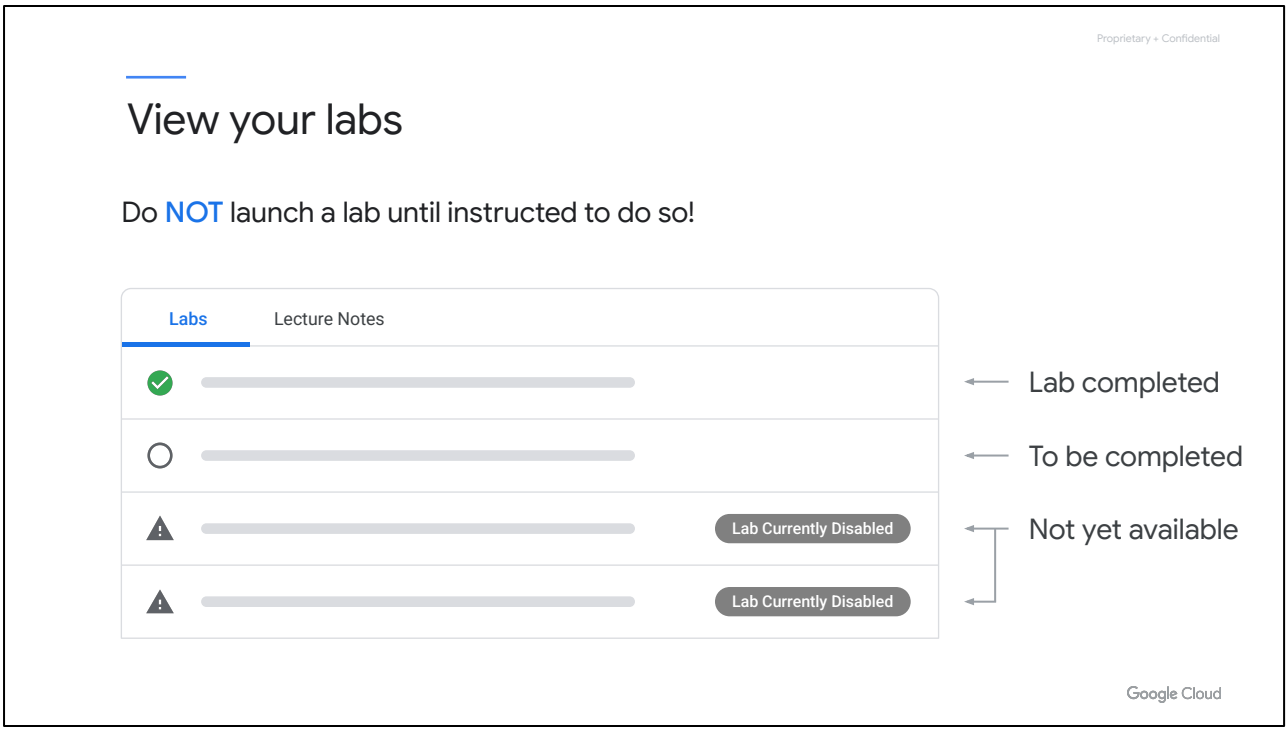

After you launch the course, you can view your labs. The lab list will indicate if a lab is:

- Completed (by you)
- Active
- Or not yet available

Your instructor will let you know when it’s time to launch a lab. Once you start a lab, you won't be able to pause and restart it, so you'll need a continuous block of time to complete the work.

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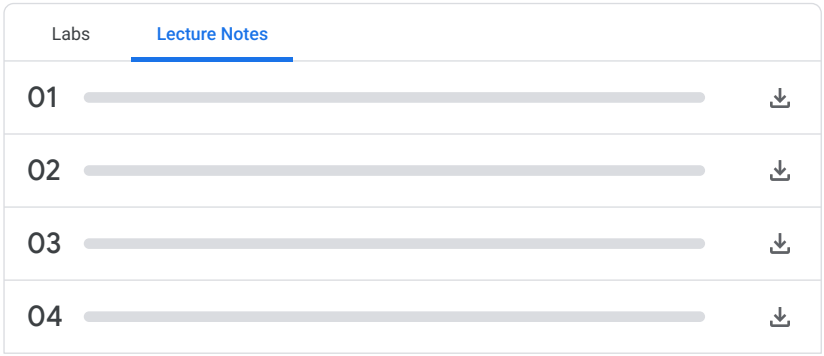
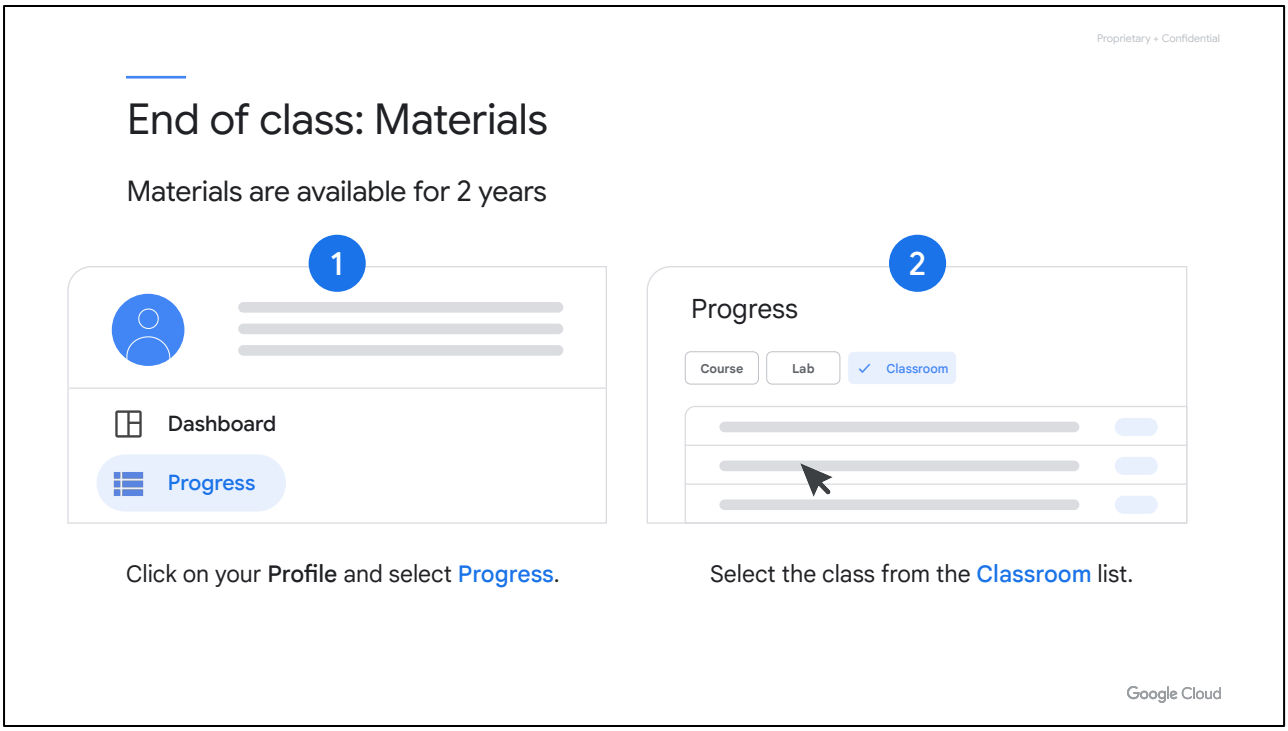
View lecture notes



You can download these as PDF files

Google Cloud

Within the course, you can also view the lecture notes. You can download these as PDF files.



You can view the course materials within Google Skills as follows:

1. Click on ***My account*** (top-right of the screen).
2. Select ***Progress***.
3. Select the ***Classroom*** activity filter.
4. Select the class from the ***Classroom*** list.

Materials are available for 2 years following the completion of a course.

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Google Cloud